Optimisation of the International Baccalaureate Diploma Programme Examination Timetable

Antony Furlong

August 2024

Supervised by Prof. dr. ir. D. den Hertog

Thesis submitted for the degree of
Master of Business Administration in Big Data and Business Analytics

Amsterdam Business School

© International Baccalaureate Organization 2024

Abstract

Producing a suitable examination timetable that adheres to certain rules and minimises stress for students is a perennial challenge for The International Baccalaureate (IB), like many other education organisations. Currently, its creation is a heavily manual and time-consuming process and the IB has little insight into the timetable's effectiveness. This study seeks to remedy this by exploring how mixed-integer linear optimisation can be employed so that the IB can automatically create such examination timetables. This would also allow the IB to investigate the impact of changes in policy in this area, such as shortening or lengthening the window of time in which examinations take place.

Once the rules and the aim of the examination timetabling problem had been fully established, the problem was first expressed as a mixed-integer linear optimisation problem. Here, the objective function was expressed as a sum of penalties relating the proximity of each pair of exams, weighted by the number of students taking that pair of exams and the constraints were based on the rules decided by IB staff. The model was then written into Python language using the Gurobi solver. Because of the expected challenges relating to computation time, the model was then tested on a small, synthetic, toy dataset to prove its functionality before being test on real IB data from a recent examination session.
The results from the toy data showed that the model could be used to create timetables, although computation time was intensive. It was also found that good, but not fully optimal solutions could be found within a few seconds.

Whilst the model was solved on the toy dataset, computation time was too extensive for real IB data and the solver had to be terminated manually after several hours. It was, however, able to find good solutions within an hour. Three potential amendments to the model, aimed at reducing the computational load, were tested on the toy data and the most promising applied to the real IB data. Whilst this showed improvements in computational speed, it wasn't sufficient for the solver to fully run. An interesting finding was that the timetables produced by both models group exams in cognately similar subjects and 'frontload' the earlier part of the week. Further potential improvements to the model and future avenues for the study are also discussed within the study.

Statement of Originality

This document is written by Antony Furlong who declares to take full responsibility for the contents of this document.

I declare that the text and the work presented in this document are original and that no sources other than those mentioned in the text and its references have been used in creating it. I have not used generative AI (such as ChatGPT) to generate or rewrite text.

UvA Economics and Business is responsible solely for the supervision of completion of the work and submission, not for the contents.

Acknowledgement

I would like to thank Prof. dr. ir. Dick den Hertog and Prof. Julien Grand-Clément for their generous, patient and insightful support throughout the entire study. This study would not have been possible without their time, their ideas and their constructive criticism. I would also like to thank my colleagues at the International Baccalaureate, not only for providing me with the data and the time needed to make this possible, but more importantly for allowing me to approach an important topic like this with total freedom.

Table of Contents

Abstract i
Statement of Originality ii
Acknowledgement ii
1. Introduction 1
1.1 Background to the problem 1
1.2 The International Baccalaureate Diploma Programme and its examination timetable 2
2. Literature review 3
3. Method 6
3.1 The IB DP examination timetable 7
3.2 Mixed integer optimisation formulation 8
4. Case study on a reduced dataset 11
4.1 Toy model creation 12
4.2 Results 12
5. IB case study 16
5.1 Data preparation and parameter selection 16
5.2 Results 18
6 Improvements 20
6.1 Model adjustments 20
6.2 Impact on solve time 21
6.3 Other possible improvements 23
7. Conclusion 25
7.1 Limitations of the study 25
7.2 Future work 26
7.3 Summary 27
References 28
Appendix 1 – Detail of the toy model 31
Appendix 2 – Recoding of IB exams 38
Appendix 3 – IB exam timetable at the point of manual termination 41
Appendix 4 – IB timetable at point of termination with amended model 44

1. Introduction

1.1 Background to the problem

A perennial challenge that the International Baccalaureate, along with many other educational organisations, faces is trying to create a timetable for when these exams should take place. Whilst there are clear start and end dates to the examination window, there are 64 (sometimes paired or combined) exams spread across just over three weeks which means that it is inevitable that exams in more than one subject need to take place at the same. This leads to the issue that students who are affected by this will need to take one of the two exams in a later free slot, which is an obvious academic integrity risk and causes stress to students and schools alike. In addition, given the numerous possible subject combinations that any student can study for, it is possible that some students will need to take multiple exams in a short space of time, leading to a further increase in stress and possible lower performance.

The IB would like to create an exam timetable that minimises the amount of additional stress the timetable could cause students by maximising the spread of exams across the timetable for as many students as possible. It will also need to adhere to a number of other pre-determined rules, such as scheduling exams in the same subject on consecutive days and a maximum number of exam clashes (see Section 3.1). In addition, the IB has recently committed to releasing the timetable two years in advance of the taking place. This is before candidate numbers are known and so exact figures will not be possible.

Currently, the examination timetable is put together manually and is a time-consuming and difficult thing to do. In addition, the IB currently has no insight into the extent to which the timetable achieves its aim of spreading exams out for students as much as possible. Furthermore, there is currently no way of assessing the impact that each of its rules has on that aim and therefore whether adjusting or removing the rule would result in a better trade-off between spread and other factors. This study would seek to change this by developing a prototype that automatically creates examination timetables that optimize fairness, that minimizes the way in which pairs of exams are scheduled closely together, keeps exam clashes to an acceptable number and adheres to other pre-determined rules. Ideally the IB would like for the prototype to produce a number of possible timetables that adhere to the constraints which the team that finalise the timetable can choose from. Longer-term the IB would like to integrate a fully operational tool that will not only provide optimised examination timetables for each examination session (two per year), but will also allow the IB to evaluate the impact that any proposed changes in examination rules (either new rules or adjustments of existing

rules) and policies in this area would have on the overall effectiveness of the examination timetable.

As such, the central research question for this study is

Can the IB's examination schedule be created through optimization techniques in such a way that maximises the spread of examinations for students whilst complying to all constraints?

1.2 The International Baccalaureate Diploma Programme and its examination timetable

The International Baccalaureate (IB) Diploma Programme (DP) is an educational offering aimed at pre-university students across the globe. In May 2024, there were 192,867 students registered for assessment in 3323 schools in 152 countries (IBO, 2024). In total, just over 2.74 million assessments are taken, of which just over 1.63 million are exams.

The Diploma Programme comprises subjects organised into six subject groups and a mandatory core. The six groups are studies in language and literature, language acquisition, individuals and societies, sciences, mathematics and the arts. Students must take at least one subject from each of the first five subject groups but may take a second subject from one of these five groups instead of an arts subject (IBO, 2014).

Most subjects are available at both higher and standard level. The higher level (HL) version of the subject is intended to be 'broader, deeper and more challenging in scope' (IBO, 2017, p2) than their standard level (SL) counterpart and is allocated 240 hours' teaching time, compared to 150 hours for SL. It is important to note that the two curricula will generally share content and there will often be common examinations or shared questions across the HL and SL exams within a subject, which has an impact on examination timetabling because these exams will need to be taken at the same time.

All subjects in the DP, with the exception of the core and arts subjects, comprise both coursework assessment and examinations and typically, there are either two or three exams per subject. All exams for a subject are taken at the end of the course's completion (i.e. the subjects are not modular). There are therefore two examinations sessions per year, taken in May and November and they typically last just over three weeks. The May session has a much larger candidature than November and so will be the focus of this study.

The timetable is currently constructed manually, with the aim of spreading out the exams for students as best as possible whilst also keeping the number of exam clashes (where a student has two exams scheduled at the same time) small. However, because this is a manual process, it is hard to evaluate how successful it is, other than through feedback from schools. Furthermore, 'spread' is enforced by trying to spread out large entry exams as much as possible across the examination window, without consideration of how many students take each combination of those exams. It is estimated that the process takes around two days to complete, with some additional time for feedback and amendments.

2. Literature review

The impact of exam scheduling on student performance is not heavily researched. Goulas & Megalokonomou (2020) analysed the impact of exam scheduling on student performance in a high school in Greece, specifically looking at the impact of the number of days between exams, the number of days since the exam schedule has started that an exam was scheduled on and the order of exams within the schedule. They found that time between exams had a negative effect on results for non-STEM (Science, Technology, Engineering and Maths) subjects but not on STEM subjects, that the length of time since the student's first exam had a negative impact on outcomes (due to fatigue) for STEM subjects (and not non-STEM subject), and that the number of previously completed exams has a positive impact on STEM subjects. Furthermore, they found that these impacts were not the same across the ability range. For instance, the impact of the number of previously completed exams having more of an impact on lower ability students than higher ability students.

Fillmore & Pope (2012) studied the impact that the Advanced Placement (AP) exam timetable had on performance and found an almost linear effect for how time between pairs of exams had an impact on performance. They also found that this effect was different for different demographic groups. Schmidt et al. (2022) found that changing an exam schedule from being spaced out to back-to-back (with a 15 minute gap) between them had a negative impact on performance in the case of two related courses at a medical school. Interestingly, this paper also considers the impact of the change of student satisfaction, which saw a similar drop. Jayakumar et al. (2016) provide one of the few studies looking at the impact of the exam schedule on well-being and find that students at a medical college consider that scheduling exams on a Friday at a single medical school was the most preferred day and Monday the least preferred.

It is important to note that the context of these papers differ from the IB as they relate to a single institution or country and so the same results might not be seen in an international context such as the IB. They also don't consider how different elements of an exam (such as question type or skill level required) might have on the impact of an exam schedule, which could be of importance to this study, given the IB's focus on higher-order thinking over memorisation of facts (IBO, 2019). That notwithstanding, it is clear that the scheduling of exams is an important factor in student performance.

Optimisation of examination timetables is a well-considered topic within the broader field timetabling optimisation, with much literature on the subject. In a detailed discussion of examination tabling, Qu et al. (2009) define such a problem as 'assigning a set of exams into a limited number of timeslots (time periods) and rooms (of certain capacity), subject to a set of constraints.'. These constraints fall into one of two categories: those that must be adhered to ('hard constraints') and those that are desired but can be relaxed if needs be ('soft constraints'). Whilst there are a wide range of different problems possible within this definition, one key difference with the IB's situation is that there is no need to consider how it allocates exams to rooms and adherence to room capacity, because the schools with students registered for the exams are expected to have sufficient capacity to host examinations.

Much of the literature on examination timetabling focusses on the context of university examinations. Carter et al. (1996), introduce the so-called 'Toronto' datasets, which were a range of real-world datasets from universities around the world with the aim of creating timetables for them which maximises the spread of exams for students whilst ensuring that there are no exam clashes (students having two exams scheduled at the same time) and adherence to other typical constraints, such as scheduling certain exams in 'preferred' periods. This paper led to a large number of research studies in this area, which were surveyed by Qu et al. (2009).

The examination timetabling track of the 2nd International Timetabling Competition has provided the basis for multiple further studies. The original mathematical optimisation formulation of the problem was non-linear (McCollum et al., 2007), however a linear version was later provided (McCollum et al., 2012). The model's objective function is a weighted sum of soft constraints multiplied by assigned penalties. The soft constraints include instances such as 'two exams in a day' and 'period spread' (instances where a student sits two exams within a given number of periods of one another), which are aimed at ensuring that the exams are spread out for students, much like the aim of this study. However, there are a number of

additional constraints which do not apply to the IB's context, such as constraints relating to room usage and exams of different durations being schedules in the same room and the same time. Another key difference with this study is that IB exams are nested within subjects and it is usually the subjects whose exams have constraints placed on them (such as no Arabic or Hebrew Language and Literature exams on a Friday afternoon), rather than individual exams being constrained in some way. Furthermore, many of the studies have a 'no clashes' constraint, whereas the IB tries to keep that below a certain number.

Many subsequent articles apply and adapt the model created by McCollum et al. (2012) to their own context or to suggest improvements on the model itself or how it was solved. Extensions to the model include the addition of further constraints relating to rooms, such as the exam taking place in a room different from the class taking place (Müller, 2016) and consideration of different examinations sharing certain questions and so needing to be scheduled together (Bazari et al., 2023) and consideration of constraints relating to exam invigilators (Mohmad Kahar & Kendall, 2014) . Siew et al. (2024, p41483) include tables that document 38 major constraints found in the literature associated with examination timetabling (tables 6 and 7). Of these, 19 relate to either room constraints or invigilator constraints and so are not relevant to the context of the study.

The previous article (Siew et al., 2024) is a survey of examination timetable of the different methods used to solve examination timetable problems from studies between 2012 and 2023. The authors conclude that there are five broad categories of solution methodologies: mathematical optimisation, heuristics, matheuristic, hyper-heuristic, metaheuristic and hybrid approaches. A key finding from the study is that mathematical optimisation approaches are popular when used on real-world datasets, however they note that this is potentially because of the relatively small size of the scenarios encountered and that they become inefficient as the problems become large-scale. The also summarise (tables 3 and 4, p41482) the size of the datasets associated with both the 2nd International Timetabling Competition (McCollum et al., 2007) and that of another dataset from Yeditepe University, first presented by Ozcan & Ersoy (2005). The number of exams in the datasets in the 2nd International Timetabling Competition range between 78 and 1096 (mean of 586.08) and the number of possible periods to assign them to range between 12 and 80 (mean of 38.67). For the Yeditepe University datasets, the number of exams range between 26 and 210 (mean of 132.38) and the number of periods range between 2 and 7 (mean of 5.23). As is shown in Section 5.1, the IB has significantly fewer exams than many instances in both sets of datasets but a similar number of available periods to the first set of datasets and more than the latter. Overall, given the relatively small number of exams and the IB not having additional room related constraints, it is reasonable to

conclude that the IB's timetabling problem is relatively small in size compared to these datasets and so mathematical optimisation approaches could be appropriate.

An under-researched area within examination timetabling is that of fairness, which is an important and highly complex topic within the broader assessment and educational field. Muklason et al. (2017) produced the first article to explicitly discuss fairness and examination timetabling. They began by surveying students about their preferences relating to examination timetables and discovered that students preferred their exams to be as spread out as possible, preferred afternoon slots (followed by morning and finally evening), that Tuesday-Thursday were their preferred days and that they preferred longer gaps before 'harder' exams. They then adjusted the standard university timetabling model by introducing a second objective function that sought to minimise the variance of the penalty scores across each student. They also adjusted their penalties to account for the relative 'difficulty' of each exam.

This approach is not applicable to the IB's context because it assumes that there is a single (and correct) value of difficulty that is the same for all students taking an exam (i.e. all students have the same view about every exam's relative difficulty). Both intersubject comparability and fairness within assessment are extremely complex topics, see Newton (2012) for a discussion on the former and Nisbet & Shaw (2019) for the latter, particularly for an international organisation like the IB and so would represent an even more complex challenge, which is beyond the scope of this study.

3. Method

For this study, the first stage is to establish what the 'rules' and aims are for the IB's examination timetable. Whilst there are some published guidelines for what the IB is trying to achieve and what its 'rules' are for its exam timetable (IBO, 2017, p144-145), these are over six years old at the time of this thesis. Furthermore, the rules were designed under the constraint that the construction of the timetable would be manual. These rules will be outlined in Section 3.1.

The next stage will be to formulate the aims and rules of the IB's examination timetable as a mixed-integer linear optimisation model (Section 3.2). Because of the possible time and computational issues with using mathematical optimisation on large problems (Siew et al., 2024), the model will then be tested on a synthetic, toy dataset to show that it can solve such a problem in Section 4. The full IB dataset will then be prepared and the model will be tested on it in Section 5 before ways to make the model more efficient are considered and tested in

Section 6. Finally, the study will be summarised and avenues for future research will be discussed in Section 7.

3.1 The IB DP examination timetable

After discussions with senior leaders within the IB's assessment division, it has been agreed that the overall aim for a fully optimised examination timetable is trying to spread examinations out for students as much as is possible whilst still adhering to the following rules:

- The examination window begins at the end of the last week in April. The only subjects allowed to have examinations on this day are School Based Syllabus and Pilot subjects.
- No examinations may take place at the weekend.
- No examinations may take place on May 1st.
- No examinations in Arabic or Hebrew Literature, Language and Literature or Language Acquisition may take place on a Friday afternoon.
- The maximum total length of exams a student may take in a single day is six hours.
- The maximum duration of any examination in an afternoon is 2 hours 15 minutes.
- Examinations in Business Management, History and English Language and Literature must take place in the first half of the examination window.
- All exams within the same subject must take place on consecutive days.
- The maximum upper limit for exam clashes (the number of situations where a student has two exams scheduled at the same time) is less than the current situation (9639 in May 2024).
- The maximum duration of the examination window is three and a half weeks.

It is important to note that these are slightly different from the rules used to create the existing exam schedules, particularly the duration of the exam window, which is slightly longer under these set of rules. Furthermore, the rule relating all exams in the same subject being on consecutive days is currently interpreted as treating Friday and the following Monday as 'consecutive days' (i.e. ignoring the weekend since no exams are allowed anyway), but this will be treated as non-consecutive for the purpose of this study. This is because the spirit of this rule is that exams within the same subject aren't on the same day because then any student who is unable to take exams on a particular day (e.g. because they are ill) doesn't miss every exams in that subject. However, they should only have to 'revise once' for each subject and therefore putting them as close together and not separating them with a weekend is most closely aligned with the spirit of the rule.

3.2 Mixed integer optimisation formulation

The following shows the mixed integer linear optimisation formulation of the situation outlined above.

3.2.1 Sets and parameters

I,J: Set of all exams (indexed as i,j=1,2,…,n).

P,Q: Set for the possible slots each day (p,q=1 indicates the morning slot, p,q=2 indicates the afternoon slot).

M,N: Set of all possible days (indexed as m=1,2,…,o where m indicates the first slot chronologically and o indicates the last one).

D_m: Day of the week to which member m refers to.

S_i: The subject to which exam i relates to

T_m: The date to which each day m relates to

$C_{i,j}$: The number of students sitting both exams i and j where $i<j$.

W^a: The penalty score to assign if two exams are scheduled in the same slot.

W^b: The penalty score to assign if two exams are scheduled in different slots on the same day.

W^c: The penalty score to assign if two exams are scheduled on consecutive days.

W^d: The penalty score to assign if two exams are scheduled with a gap of one day.

W^e: The penalty score to assign if two exams are scheduled with a gap of two days.

W^f: The penalty score to assign if two exams are scheduled with a gap of three days.

W^g: The penalty score to assign if two exams are scheduled with a gap of four days.

L_i: The length of exam i in hours.

3.2.2 Variables

$x_{i,p,m}$: Binary variable representing the allocation of exam i to slot p of day m on the timetable (a value of 1 indicates that it is and 0 if it isn't).

$y^a_{i,j}$: Binary variable representing whether or not exams i and j are assigned to the same slot in the timetable (a value of 1 indicates that they are and 0 indicates that they aren't). For i,j in I,J such that $i<j$.

$y^b_{i,j}$: Binary variable representing whether or not exams i and j are assigned the same day but in a different slot (i.e. one in the morning, the other in the afternoon). For i,j in I,J such that $i<j$.

$y^c_{i,j}$: Binary variable representing whether or not exams i and j are assigned to slots in consecutive days. For i,j in I,J such that $i<j$.

$y^d_{i,j}$: Binary variable representing whether or not exams i and j are assigned to slots with one day between them. For i,j in I,J such that $i<j$.

$y^e_{i,j}$: Binary variable representing whether or not exams *i* and *j* are assigned to slots with two days between them. For *i,j* in I,J such that *i<j*.

$y^f_{i,j}$: Binary variable representing whether or not exams *i* and *j* are assigned to slots with three days between them. For *i,j* in I,J such that *i<j*.

$y^g_{i,j}$: Binary variable representing whether or not exams *i* and *j* are assigned to slots with four days between them. For *i,j* in I,J such that *i<j*.

3.2.3 Objective function

The following objective function is used to maximise the spread of examinations for students.

$$\sum_{i<j\in I,J} C_{i,j}(W^a y^a_{i,j} + W^b y^b_{i,j} + W^c y^c_{i,j} + W^d y^d_{i,j} + W^e y^e_{i,j} + W^f y^f_{i,j} + W^g y^g_{i,j}) \qquad (3.1)$$

This is a sum of all penalties applied to all pairs of examinations *i* and *j*, weighted by the number of students taking both exam *i* and exam *j*. The structure and penalties are similar to that used in Muklason et al. (2017) without the additional penalties relating to room allocation, front-loading of large entry examinations or number of slots having examinations in them, since these do not apply to the IB's situation.

3.2.4 Constraints

The first set of constraints are not related to the IB-specific rules set out in Section 3.1 but describe general aspects of the examination timetable

$$\sum_{m\in M}\sum_{p\in P} x_{i,p,m} = 1 \qquad \forall i \in I \qquad (3.2)$$

$$x_{i,p,m} \in \{0,1\} \qquad \forall i \in I \qquad (3.3)$$

$$y^a_{i,j} \geq x_{i,p,m} + x_{j,p,m} - 1 \qquad \forall p \in P, \forall m \in M \text{ with } m=n, \forall i,j \in I \text{ with } i<j \qquad (3.4)$$

$$y^b_{i,j} \geq x_{i,1,m} + x_{j,2,m} - 1 \qquad \forall m \in M, \forall i,j \in I \text{ with } i<j \qquad (3.5)$$

$$y^b_{i,j} \geq x_{i,2,m} + x_{j,1,m} - 1 \qquad \forall m \in M, \forall i,j \in I \text{ with } i<j \qquad (3.6)$$

$$y^c_{i,j} \geq \sum_{p,q\in P}(x_{i,p,m} + x_{j,p,m+1} - 1) \qquad \forall m \in M \text{ with } 1\leq m\leq O-1, \forall i,j \in I \text{ with } i<j \qquad (3.7)$$

$$y^c_{i,j} \geq \sum_{p,q\in P}(x_{i,p,m+1} + x_{j,p,m} - 1) \qquad \forall m \in M \text{ with } 1\leq m\leq O-1, \forall i,j \in I \text{ with } i<j \qquad (3.8)$$

$$y^d_{ij} \geq \sum_{p,q\in P}(x_{i,p,m} + x_{j,q,m+2} - 1) \qquad \forall m \in M \; 1\leq m\leq O-2 \qquad \forall i,j \in I \text{ with } i<j \qquad (3.9)$$

$$y^d_{ij} \geq \sum_{p,q\in P}(x_{i,p,m+2} + x_{j,q,m} - 1) \qquad \forall m \in M \; 1\leq m\leq O-2 \qquad \forall i,j \in I \text{ with } i<j \qquad (3.10)$$

$$y^e_{ij} \geq \sum_{p,q\in P}(x_{i,p,m} + x_{j,q,m+3} - 1) \qquad \forall m \in M \; 1\leq m\leq O-3 \qquad \forall i,j \in I \text{ with } i<j \qquad (3.11)$$

$$y^e_{ij} \geq \sum_{p,q\in P}(x_{i,p,m+3} + x_{j,q,m} - 1) \qquad \forall m \in M \; 1\leq m\leq O-3 \qquad \forall i,j \in I \text{ with } i<j \qquad (3.12)$$

$$y^f_{ij} \geq \sum_{p,q\in P}(x_{i,p,m} + x_{j,q,m+4} - 1) \qquad \forall m \in M \; 1\leq m\leq O-4 \qquad \forall i,j \in I \text{ with } i<j \qquad (3.13)$$

$y_{ij}^{f} \geq \sum_{p,q \in P}^{\square}(x_{i,p,m+4} + x_{j,q,m} - 1)$ $\forall m \in$ M 1≤m≤O-4 $\forall i,j \in$ I with $i<j$ (3.14)

$y_{ij}^{g} \geq \sum_{p,q \in P}^{\square}(x_{i,p,m} + x_{j,q,m+5} - 1)$ $\forall m \in$ M 1≤m≤O-5 $\forall i,j \in$ I with $i<j$ (3.15)

$y_{ij}^{g} \geq \sum_{p,q \in P}^{\square}(x_{i,p,m+5} + x_{j,q,m} - 1)$ $\forall m \in$ M 1≤m≤O-5 $\forall i,j \in$ I with $i<j$ (3.16)

$y_{i,j}^{a},\ y_{i,j}^{b}, y_{i,j}^{c}, y_{i,j}^{d}, y_{i,j}^{e}, y_{i,j}^{f}, y_{i,j}^{g} \in [0,1]$ $\forall i,j \in$ I with $i<j$ (3.17)

Constraint (3.2) ensures that each exam is allocated to exactly one slot and (3.3) ensures each $x_{i,p,m}$ is a binary variable. Constraints (3.4) to (3.14), along with (3.17) describe the relationships between the various $y_{i,j}$ variables and their $x_{i,p,m}$ counterparts. Using (3.4) as an example, if $x_{i,p,m}$ and $x_{j,p,m}$ are not both equal to 1, then the minimisation of $y_{i,j}^{a}$ in the objective function forces it to be equal to 0. When both $x_{i,p,m}$ and $x_{j,p,m}$ are equal to 1, $y_{i,j}^{a}$ must be equal to 1. The remaining constraints from (3.5) to (3.14) follow in a similar fashion. It is important to note that both equations relating to each $y_{i,j}$ (e.g. (3.6) and (3.7)) are needed because $i<j$ has been specified (i.e. separate constraints are needed to account for exam i being the day before or the day after exam j).

Constraint (3.17) defines that $y_{i,j}$ as a continuous variable with limits of 0 and 1. Whilst this might appear to be a contradiction with how it is defined in Section 3.2.2, constraints (3.4)-(3.17) mean that the only values they can take are 0 and 1. The reason for this formulation is in order to reduce the processing time of the algorithm and to save memory.
The constraints below describe each of the IB-specific constraints in order as laid out in Section 3.1.

Exams in school-based syllabus (SBS) and pilot subjects (and only those subjects) take place on the first day of the examination window

$x_{i,p,m} = 0$ for m=1 $\forall i \in$I with $S_i \notin$ {SBS}, $\forall p \in$ P (3.18)

$\sum_{p \in P}^{\square} x_{i,p,m} = 1$ for m=1 $\forall i \in$I with $S_i \in$ {SBS} (3.19)

No examinations may take place at the weekend

$x_{i,p,m} = 0$ $\forall m \in$ M with $D_m \in$ {Saturday, Sunday}, $\forall p \in$ P (3.20)

No examinations may take place on May 1st

$x_{i,p,m} = 0$ $\forall m \in$ M with T_m=May 1st, $\forall p \in$ P (3.21)

No examinations in Arabic or Hebrew Literature, Language and Literature or Language Acquisition may take place on a Friday afternoon

$x_{i,2,m} = 0$ $\forall i \in$ I with $S_i \in$ {Arabic Literature, Hebrew Literature, Arabic Language and Literature, Hebrew Language and Literature, Arabic Language Acquisition, Hebrew Language Acquisition} and $\forall m \in$ M with $D_m \in$ {Friday} (3.22)

The maximum total length of exams a student may take in a single day is six and a quarter hours

$y^b_{i,j} = 0 \qquad \forall i,j \in I \text{ such that } (L_i + L_j) > 6.25 \text{ and } i < j$ (3.23)

The maximum duration of any examination in an afternoon is 3 hours

$x_{i,2,m} = 0$ $\forall i \in$I with $L_i > 3$, for $\forall m \in$ M (3.24)

Examinations in Business Management, History and English Language and Literature must take place in the first half of the examination window

$\sum_{m \in M/\{0,o/2\}} \sum_{p \in P} x_{i,p,m} = 0$ $\forall i \in$I where $S_i \in$ {Language and Literature, Business Management, History}, $\forall p \in P$ (3.25)

This constraint forces $x_{i,p,m}$ to be 0 for all slots in the second half of the exam timetable, which alongside constraint (3.2) forces these exams into the first half of the timetable. The double summation here is used to minimise the total number of constraints.

All exams within the same subject must take place on consecutive days

$\sum_{p \in P} x_{i,p,m} + \sum_{q \in P} \sum_{n \in M/\{m-1.m+1\}} x_{j,q,n} \leq 1$ $\forall$ $i,j \in$I where $i<j$ and $S_i = S_j$ $\forall m \in$ M (3.26)

Here, $x_{j,q,n}$ is being forced to be 0 whenever $x_{i,p,m}$ is 1 and n is not m-1 or m+1. Therefore, this forces $x_{j,q,n}$=1 in one of n=m-1 or n=m+1, since $x_{j,q,n}$ must equal 1 exactly once.

The maximum upper limit for exam clashes is lower than the current situation (9369 in May 2024)

$\sum_{i<j \in I} C_{i,j}\, y^a_{i,j} \leq 9368$ (3.27)

4. Case study on a reduced dataset

Since this model in the context of the IB (64 exams over three and a half weeks) is rather complex and likely to be computationally challenging, the first stage is to test the model and its feasibility with a smaller, toy dataset before moving on to the full IB dataset.

4.1 Toy model creation

For this initial case study, the data was completely engineered and created in such a way to prove that the entire model that would be used on IB data runs properly and solves such a problem, but without the expected concerns around computational load.

The synthetic data included 20 exams, with associated subjects and exam lengths so as to match IB data as far as is possible. For every constraint that includes a subject name, there was at least one exam associated with a subject within that constraint and so all constraints were included. For the constraint enforcing exams in certain subjects to take place in the first half of the examination window, the subjects for which this applied to was chosen to be 'Finance' and 'Law and Ethics'.

In addition, a matrix of synthetic counts of students taking pairs of exams ($C_{i,j}$ values) was also created with values ranging from 0 to 100. The maximum duration of the examination window was shortened by a week to 16 days and the number of slots in a day was kept at two (morning and afternoon). The W^a, W^b etc. values were set in in a similar fashion to those found in Muklason et al. (2017), but the penalty relating to exams being furthest apart but still getting a penalty (W^g) was set at 1 and then the value doubled for each time the W represented exams being closer together, i.e. W^f was set at 2, W^e was set at 4, etc. Because of the relatively small $C_{i,j}$ values in comparison to the real-life data, the maximum acceptable number of exam clashes was set at 15.

The full list of exams, their subjects and exam lengths can be found in Appendix 1.1 and the $C_{i,j}$ values in appendix 1.2.

The model was encoded using Python 3.9.13. Specifically, the model was created using the gurobipy package and solved with Gurobi Optimizer version 11.0.2. The code can be found in appendix 1.3. The program was run on a Macbook Air with an M1 chip and with 16GB of installed memory.

4.2 Results

As expected, solving even the toy dataset was quite computationally intensive. In total, the algorithm took 482.98 seconds to solve the optimization problem and explored 253,341 nodes with 18,573,043 iterations. The Gurobi optimizer also shows the number of 'work units' taken, which was 685.62. Work units show how much work is done to solve a given

optimization problem, but unlike runtime, will be the same every time you run the same model using the same hardware ("Work," n.d.).

Given the time taken, it is interesting to note that the best objective value (25,910) was found after 25 seconds, with the remainder of the time spent 'verifying' that this solution is optimal. In addition, solutions with an objective value very close (within 10%) to the final objective value was found after three seconds (see table 1). Therefore, it also found very good, if not optimal, solutions relatively quickly.

Time elapsed (s)	Objective value of solution	Percent deficit to optimal value	Gap
0	31,440	21.34%	70.8%
1	29,410	13.51%	67.9%
2	29,010	11.96%	67.4%
3	27,860	7.52%	66.0%
23	27,550	6.33%	55.8%
24	26,910	3.86%	54.8%
25	25,910	0%	51.3%

Table 1 – Progression of the objective function

Figure 1 shows how the 'gap' progressed during the duration of the process. The 'gap' is the distance between the objective value at that time and the lower bound expressed as a percentage of that objective value. It shows that the decline was fairly steady after around 200 seconds and that it hadn't got close to zero quickly.

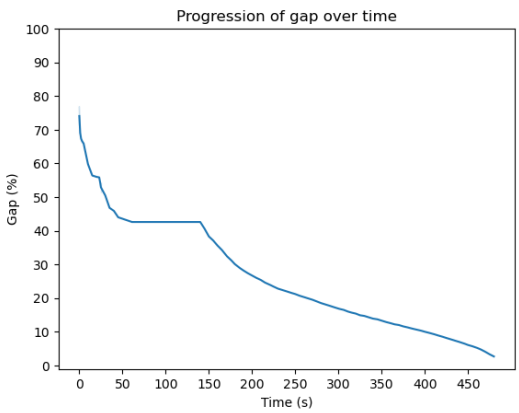

Figure 1 – Progression of the 'gap' over time

The final timetable for this dataset is shown in table 2. It is clear from this that the model 'worked' in that it produced the optimal solution that satisfied all the constraints. It is interesting to note that the solution doesn't include exams on every available day, since Wednesday 6th May is left clear.

Day of Week	Date	Slot	Exam Name
Thursday	23/04/2026	AM	SBS Exam 1
Friday	24/04/2026	AM	SBS Exam 2
Monday	27/04/2026	AM	Optimisation Exam 2
Monday	27/04/2026	AM	Law and Ethics Exam 2
Monday	27/04/2026	PM	Machine Learning Exam 1
Tuesday	28/04/2026	AM	Machine Learning Exam 2
Tuesday	28/04/2026	PM	Optimisation Exam 1
Tuesday	28/04/2026	PM	Law and Ethics Exam 1
Wednesday	29/04/2026	AM	Statistics Exam 2
Wednesday	29/04/2026	PM	Finance Exam 1
Thursday	30/04/2026	AM	Statistics Exam 1
Thursday	30/04/2026	PM	Finance Exam 2
Monday	04/05/2026	AM	Coding Lab Exam 1
Monday	04/05/2026	PM	Language A Literature Exam 2
Tuesday	05/05/2026	AM	Language A Literature Exam 1
Tuesday	05/05/2026	PM	Coding Lab Exam 2
Thursday	07/05/2026	AM	Deep Learning Exam 1
Thursday	07/05/2026	PM	Data Stewardship Exam 2
Friday	08/05/2026	AM	Deep Learning Exam 2
Friday	08/05/2026	PM	Data Stewardship Exam 1

Table 2 – The final timetable for the toy dataset

To show how the optimal model differs from merely a 'good' model, table 3 shows the exam timetable for the solution with an objective function of 27,750, which was found after 23 seconds. Those exams which have been allocated the same slot as the optimal timetable are highlighted in bold. This applies to eight of the 20 exams, with Data Stewardship Exam 3 and Deep Learning Exam 1 (in italics) being on different slots on the same day, which is of no practical difference and so are very close to being the same.

Day of Week	Date	Slot	Exam Name
Thursday	23/04/2026	AM	**SBS Exam 1**
Friday	24/04/2026	AM	**SBS Exam 2**
Monday	27/04/2026	AM	**Optimisation Exam 2**
Monday	27/04/2026	AM	Finance Exam 2
Monday	27/04/2026	PM	**Machine Learning Exam 1**
Tuesday	28/04/2026	AM	**Machine Learning Exam 2**
Tuesday	28/04/2026	PM	**Optimisation Exam 1**
Tuesday	28/04/2026	PM	Finance Exam 1
Wednesday	29/04/2026	AM	Law and Ethics Exam 1
Thursday	30/04/2026	AM	Law and Ethics Exam 2
Monday	04/05/2026	PM	Statistics Exam 2
Tuesday	05/05/2026	AM	Statistics Exam 1
Tuesday	05/05/2026	AM	Language A Literature Exam 1
Tuesday	05/05/2026	PM	Coding Lab Exam 2
Wednesday	06/05/2026	AM	Language A Literature Exam 2
Wednesday	06/05/2026	PM	Coding Lab Exam 1
Thursday	07/05/2026	AM	*Data Stewardship Exam 2*
Thursday	07/05/2026	PM	*Deep Learning Exam 1*
Friday	08/05/2026	AM	**Deep Learning Exam 2**
Friday	08/05/2026	PM	**Data Stewardship Exam 1**

Table 3 – An example 'good', but not optimal, timetable

Structurally, this timetable seems very similar to the optimal timetable, with the only significant differences being that there Wednesday 6th May is no longer empty but the second half of the first full week is less busy, with just the 'Law and Ethics' exams taking place then, rather both than 'Statistics' and 'Finance'.

In summary, this has shown that the model and its coding in Python can be used to solve problems of this nature with toy datasets. However, it seems to be computationally intensive; the extent to which this is an issue will be explored in the next section where full-scale IB data is used. There is however, some evidence that a good solution can be found quickly, as evidenced by the optimal solution being found after 23 seconds and a 'good' solution found after 3 seconds.

5. IB case study

The previous section showed that the model outlined in Section 3 can solve problems with small datasets, although the computation time is considerable. The next step is to apply this to IB data.

5.1 Data preparation and parameter selection

The data used was from the May 2024 examination session. This data came in its raw format as a list of assessment being taken for every (anonymised) candidate. This totalled to 29,292,23 entries, of which 1,754,938 relate to examinations. The remainder are registrations for coursework-based assessments and so removed from the dataset. All data preparation was done using R version 4.4.0 and the 'tidyverse' package.

Figure 2 shows a section from the exam schedule in May 2024 (IBO, 2024).

Tuesday 7 May			
Morning session		**Afternoon session**	
Language B HL paper 2		History HL/SL paper 1	1h
- Listening comprehension	1h	History HL/SL paper 2	1h 30m
Language B SL paper 2			
- Listening comprehension	45m		
Language ab initio SL paper 2			
- Listening comprehension	45m		
* Except: English, French & Spanish			
Latin HL paper 2	1h 30m		
Latin SL paper 2	1h 30m		

Wednesday 8 May			
Morning session		**Afternoon session**	
History HL paper 3	2h 30m	Chemistry HL paper 1	1h
		Chemistry HL paper 3	1h 15m
		Chemistry SL paper 1	45m
		Chemistry SL paper 3	1h
		Design technology HL paper 1	1h
		Design technology SL paper 1	45m
		Design technology HL/SL paper 2	1h 30m

Figure 2 – Snippet from the May 2024 IB examination schedule

Here, History HL/SL paper 1 and paper 2, which are considered as different exams in the raw dataset are sat back-to-back in the same slot and combined into a single 'exam'. The same is true of Chemistry paper 1 and paper 3 and Design Technology paper 1 and paper 2 and many other exams within the same subject. This is not true for exams in all subjects (for example, Latin paper 2 in this snippet) and which exams are combined together is decided manually and the IB doesn't store these combinations on a database. It is beyond the scope of this study to consider how exams in the same subject might be combined into one slot to optimise the timetable, so the existing combinations from the May 2024 exam session were encoded

manually as part of the data preparation phase. The length of the 'exams' in this case was the sum of the pairs of exams.

Another step of data preparation is to enforce certain exams to be taken at the same time for academic integrity reasons. For instance, in this snippet, the subjects Language B HL, Language B SL and Language ab initio SL paper 2 exams share some questions and sources and so have to be allocated the same slot, irrespective of how many candidates take both exams. This is also the situation for Literature, Language and Literature and Literature and Performance, which are all combined and is frequently the case for the same exams at different levels within the subject (e.g. Chemistry HL paper one and Chemistry SL paper one have shared content).

Whilst these could be kept as separate assessments and an extra constraint being added to enforce these exams to take place in the same slot, these were recoded manually to be a single subject. This was done in order to keep the model as compact and simple as possible, particularly given the time taken for the case study in Section 4 to be solved. In all these cases, the length of the exam is calculated to be the longest of the two (or more) exam lengths that have been combined together.

In addition, it is possible for exams both of the previous steps (sitting exams within a subject back-to-back **and** enforcing certain exams to be taken at the same time) to be happening together. For example, Chemistry HL paper 1 and paper 3 are sat back-to-back in the same slot, as are Chemistry SL paper 1 and paper 3, but the HL and SL paper 1 exams have common content and so need to be scheduled at the same time. Hence these four exams are combined into a single exam (Chemistry paper 1 and 3) with the length being 2hr 15 min (the longer of the sums of the two sets of back to back exam lengths). The full list of all recoding can be found in Appendix 2.

The equation relating to constraint (3.26) in the model (exams in the same subject must take place on consecutive days) assumes that all subjects have exams that would take only two slots. Further investigation showed that there are two exceptions to this: Mathematics Analysis and Approaches HL and Mathematics Application and Interpretation HL, which need three different slots. There currently isn't a business rule for what happens in this case but currently, the first two exams in this subject are scheduled on consecutive days and the third is fitted in wherever it can go. Therefore, the third exam (paper three) was recoded to be treated as if it was in a completely different and unique subject.

After all recoding had taken place, there were 64 exams (or combinations of exams being sat back-to-back in the same slot) to assign slots on the schedule to. Calculating the number of students who sat pairs of exams (the $C_{i,j}$ values) was done directly from the raw data. For each pair of exams, a subset was created from the raw data that only included the registrations for those two exams and then the number of students that appeared more than once in that subset was counted.

For the model to be run, the values of a number of hyperparameters had to be selected. The cost parameters were kept the same as for the toy model, however the number of days was extended to reflect the examination window that the IB would like to use. The IB publishes its exam schedules two years in advance and so, under the new rules, this would be from Thursday 25th April to Friday 15th May in 2026, so 23 days in total. As is standard, the number of slots per day was set two. The model was solved using the same code (adjusted for the new data) as for the toy model and on the same computer.

5.2 Results

With the full IB dataset, there were a total of 14,112 continuous variables, 2,944 integer (binary) variables and 544,822 constraints. Given the size of the model, the time taken to solve was too extensive to run in its entirety and the algorithm was manually terminated after 38,521.32 seconds (10.70 hours) having explored 111,453 nodes with 57,825,637 simplex iterations. The objective value at this point was 27,210,752 and the 'gap', which is the distance between the objective value at that time and the lower bound expressed as a percentage of that objective value ("MIPGap," n.d.), was 50.7. Given that the algorithm terminates when the gap is 0 and it had only dropped by 0.3% in the final hour before it was manually terminated, it is reasonable to conclude that the algorithm was not close to terminating naturally. Therefore, the incumbent solution at the point of termination is clearly not necessarily the optimal solution and, given the large 'gap', it is hard to say how close it would have been to the objective value of the optimal solution.

The timetable that was the best at the time of manual termination can be found in Appendix 3. It is interesting to note that, similar to the toy model, Wednesday is the least 'busy' day, followed by Friday (see figure 3), with considerably fewer exams scheduled on those days than on Mondays and Tuesdays. This is perhaps a result of the weights that are applied in the objective function so heavily penalising pairs of exams that are close together and the fact that weekend days can't have any exams scheduled during them. Furthermore, the constraint that exams within the same subject need to take place on consecutive days and that 1st May falls on

a Friday will both have had an impact. It is also interesting to note that cognately similar subjects tend to be scheduled close together, for instance, many science-based subjects are scheduled on Monday 11th and Tuesday 12th May. This is likely to be because of the nature of the IB Diploma, which requires students to take subjects from a range of different subject groups, meaning that the $C_{i,j}$ values for pairs of subjects within a subject group are likely to be low, relative to the number of students doing the two subjects individually, in spite of them being cognately similar.

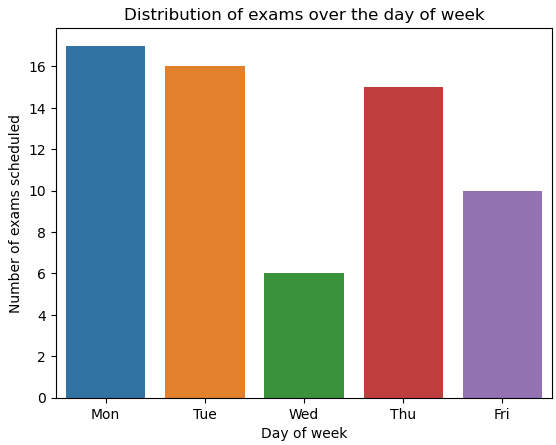

Figure 3 – Distribution of exams over the days of the week for the terminated model

It is important to remember that the current approach to examination timetable setting is manual, time-consuming (around two days) and is done without full knowledge of its real impact. Therefore, the results need to be taken in the context that it is likely to be of great benefit to the IB if a good timetable can be found relatively quickly, even if it isn't necessarily the most optimal one possible. The first feasible solution was found after 37 seconds and after an hour, the incumbent solution had an objective value of 29,713,770, which is 9.20% higher than the value at manual termination, so the remaining 9.7 hours provided an improvement of less than 10%. How much these solutions are 'better' than the currently published timetable for May 2026 is impossible to say, given the recent change in rules outlined in Section 3.1. It is, however, promising that it produces feasible timetables that comply to all the rules set out, including having fewer exam clashes than currently.

In summary, the model did produce exam timetables that adhere to the IB rules and are plausible within an hour, which is much quicker than the current, manual process. Unfortunately, however, the computation time was so large that it didn't find the optimal solution. Therefore, the next question is whether or not the model can be altered to make it

solve in a reasonable length of time, or if there are other options available. This will be discussed in the next section.

6 Improvements

This section will discuss potential ways that the model can be adjusted to bring the computation time down.

6.1 Model adjustments

Simplification of the objective function

The current objective function is as follows

$$\sum_{i<j\in I,J} C_{i,j}(W^a y_{i,j}^a + W^b y_{i,j}^b + W^c y_{i,j}^c + W^d y_{i,j}^d + W^e y_{i,j}^e + W^f y_{i,j}^f + W^g y_{i,j}^g)$$

This necessitates the creation of several variables (the $y_{i,j}$s) in order to allow for penalisation of any pair of subjects scheduled within five days of each another, but this creates a lot of complexity in the model. In total, there are 14,112 continuous variables and 2,944 integer (binary) variables when the model is run on the full IB dataset and only $y_{i,j}^a$ and $y_{i,j}^b$ needing elsewhere in the model (equations (3.27) and (3.23) respectively). The current weights (W values) are heavily tilted towards exams scheduled very close to each other and it is somewhat unavoidable that many exams with common students will be scheduled within five days of one another given the number of exams to be scheduled and the number of days available (23, of which seven are weekdays or bank holidays and two are only allowed to have SBS subjects scheduled on them). Therefore, one potential option to simplify the model would be to simplify the objective function and only penalise pairs of exams that are both taking place within a shorter time window. For instance, a model that doesn't penalise pairs of exams with more than a two-day gap between them (the equivalent of a weekend) would make the objective function

$$\sum_{i<j\in I,J} C_{i,j}(W^a y_{i,j}^a + W^b y_{i,j}^b + W^c y_{i,j}^c + W^d y_{i,j}^d) \qquad (3.28)$$

and would reduce the number of continuous variables down to 8,064.

Ordering of exams within a subject

Currently, there are no rules in place about the order exams need to take place, other than certain individual exams needing to take place (or not take place) on certain days or in the first half of the examination window. One possible change could be to enforce that exams within a

subject need to take place in a certain order, e.g. Latin paper one must take place before Latin paper two. Currently, there is no consideration of the way that exams are ordered, although current practice is in principle to put the exam named 'paper one' first. If the exams were to be taken in a specific order within a subject, this would reduce the feasible region and so could have a positive impact on computation load. In order to do this, constraint (3.26), which relates to the exams within a subject needing to be taken in consecutive days would be changed to the following

$$\sum_{p \in P} x_{i,p,m} = \sum_{q \in P} x_{j,q,m+1} \qquad \forall\ i,j \in I \text{ where } i<j \text{ and } S_i = S_j \ \ \forall m \in M \qquad (3.29)$$

Due consideration of how this change would affect the final objective value were it to be made is needed. However, it is important to also consider whether there are benefits to student wellbeing and performance by doing this that can't be measured. For instance, putting an exam that requires the least revision first could help students 'warm-up' in that subject.

Removing constraints

Another possibility would be to consider removing constraints that are not crucial to the model. One possibility could be to remove the constraint relating to capping the total number of exam clashes, constraint (3.27), if it is found to be having a big impact on computation time. The objective function is likely to keep this number low in any case and this could be tightened by increasing the weight against $y_{i,j}^{a}$. However, given that the reason behind this constraint is related to examination security, if this was removed, the number of clashes would need to be checked to ensure it is still at an acceptable level.

6.2 Impact on solve time

Any decision relating to changing the model would need to consider the trade-off between computational benefit and creating a less effective model as a result. Therefore, it is important to consider how much benefit these changes would bring. To test this each of these changes (and combination of changes) were made in turn and the model with the toy dataset was solved. Table 4 shows a summary of the time taken and work units for each of the adjusted models.

Model Adjustments	Solve time (s)	Work units	Objective value	Work units as a percentage of full model (685.62)
A	446.43	613.85	19,720	89.53
B	15.51	24.33	25,190	3.55
C	1101.54	1327.20	25,190	193.58
A&B	8.89	13.25	19,720	1.93
A&C	434.09	623.74	19,720	90.97
B&C	20.52	34.05	25,190	4.97
A&B&C	10.38	15.17	19,720	2.21

Table 4 – Summary of model performance for each of the three adjustments and their combinations. A - Reduced Objective Function, B – Ordered exams within a subject, C – Removal of maximum clash constraint

It is clear from the table 3 that simplifying the objective function and ordering the exams within a subject both had a positive impact on the speed at which the model was solved. The simplified objective function reduced the work units to 89.53% of the original model and ordering the exams within a subject reduced it to under 4%. Together, the work units went down to under 2% of the original model. In the case of ordering exams, it didn't have an impact on the overall objective function because the $C_{i,j}$s in the toy data were set up to be the same for all exams within a subject to reflect all students entered for a subject taking all exams in a subject. This is what happens in the IB, however, given the combining of exams and subjects that took place in the data preparation phase, it isn't necessarily the case that the objective value wouldn't be affected. Removing the constraint that placed a cap on the number of clashes increased the time and work units required for solving and so did not seem to improve the solve time of the model.

To see whether this would bring about sufficient improvement to make a new model operationally feasible, the model with an adjusted objective function and fixed ordering of exams within a subject were tested on the full IB dataset. The ordering of the exams within a subject was done alphabetically, given the lack of knowledge of every exam, however if this were applied live, this would be done in discussion with domain experts.

Even with the changes, the new model was still computationally intensive. The algorithm was terminated manually after slightly more than four hours (14,531.72s) of running after exploring 163,514 nodes and 75,440,414 iterations. The gap at this point was 27.4% and had only improved by 1.2% in the final hour of running. The objective value at the point of termination was 20,841,416. It is important to note that the first feasible solution was found after 14 seconds and the incumbent solution after first hour (which was actually found after 1576 seconds, or just over 26 minutes) had an objective function of 21,039,030. Therefore, for the last three and a half hours that the algorithm ran for, the improvement in the objective function was less than 1%. Therefore, whilst it hadn't come close to proving that it had found the optimal timetable, it again still found 'good' solutions within half an hour.

The timetable at the point of termination can be found in Appendix 4. It is interesting to note that there are still relatively 'quiet' days on Wednesday and Friday (see figure 4).

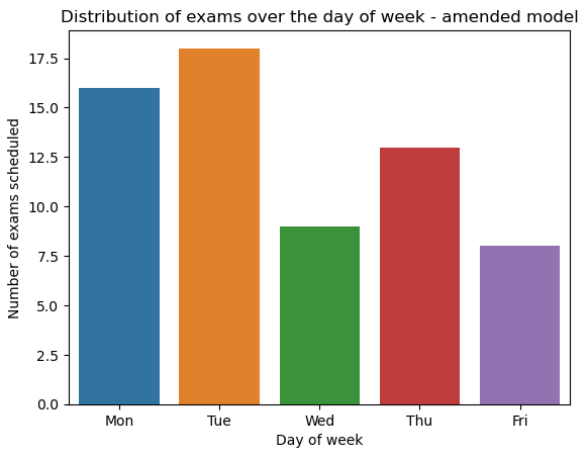

Figure 4 – Distribution of exams per day of week for the amended model

6.3 Other possible improvements

It is clear that even with these changes to the model, it is still computationally very intense and would be a challenge to use operationally. Therefore, this section discusses other options that could be considered.

Terminating the algorithm early

One possibility is simply to terminate the algorithm before their terminates naturally. Whilst this means that the optimal result is not necessarily found, it is important to consider what is meant by 'optimal' here. The weights for the objective function were taken from the literature on the subject, however there is no certainty that these are appropriate within the IB context and treats all pairs of exams as the same in terms of how much of a penalty should be applied

when they are close to one another. Whether that is valid is somewhat questionable. For instance, if exams in two subjects with some overlap in terms of the content or skills being assessed are close to each other, is this the same challenge as if two exams were in very different subjects? It is also important to note that the $C_{i,j}$ values are based on exam registrations for the exam session two years prior to one for which the timetable is being created. Therefore, they are estimates of the actual number of common candidates taking each pair of exams.

Given the computational challenges already being faced, it would be beyond the scope of this study to amend the model to account for these issues, however it does place the objective values associated with the solutions in context. Whether or not the solution with the lowest objective value is actually the 'best' solution in real-life is not certain and as such, the difference between a 'good' solution and the 'optimal' solution may be very little, if any, in practical terms.

If this was to be a way forward, a question would then be how to decide at what point to terminate the algorithm? Two options within the Gurobi solver are to terminate when the 'MIPgap' drops below a given value, or to set a time limit. Given that the MIPgap was at 50.7% and 27.4% for the two models using IB data, the change would need to be large in order for it to terminate in a reasonable time.

Terminating based on a time limit is also a possibility and perhaps a more pragmatic choice. This is likely to be a big factor in implementation, and has also been used in other exam timetabling studies, e.g Müller (2016). However, given that the toy model found the optimal solution very quickly and the objective value for both of the models run on IB data didn't improve considerably in the final hour before termination, one option could be to write a function that forces the algorithm to terminate if the objective value hasn't improved by more than a fixed percentage after a certain timeframe. Whilst this runs the risk of there being a solution that is a considerable improvement not being found, so does doing so with a termination based on time or MIPgap and this has the advantage of not terminating very early whilst improvements are still being made (as with the MIPgap) or wasted computation time (as with termination based on time).

Warm starting

Warm-starting is a technique often used for optimisation problems (e.g. Jiménez-Cordero et al., 2022) and is aimed at reducing computation time. It often does this by using a good, but

not optimal, solution as a starting point, but can also be a partial solution (AlainChabrier, 2021). If this was to be applied to this situation, the natural question is what would the starting point be?

One possibility could be to simply use the timetable from the previous year, assuming that it is close to the optimal timetable for the year it was created for. The $C_{i,j}$ values would not be the same for the year the timetable is being created for, but a bigger issue would be whether or not that timetable is feasible. The constraint around the maximum number of clashes might be violated with the updated $C_{i,j}$ values, which is fairly simply resolved by increasing the number allowed, for instance in line with the growth rate in candidature.

The other issue is that the dates of the exam window will be slightly different each year. This will mean that the constraint that excludes any exams taking place on May 1st is potentially violated because it is in a different position on the timetable each year (e.g. day 7 instead of day 5).

7. Conclusion

7.1 Limitations of the study

There were a number of limitations with this study. Firstly, it only considered one exam session in one year and so there is no certainty that these results would be replicated for other years. In particular, May 1st, where no exams are allowed (constraint (3.21)), will fall on different days of the week each year, which might have a big overall impact on the way that exams are distributed throughout the timetable. This would be particularly noticeable if it fell on a Tuesday, because without a relaxation of one of either the constraint enforcing exams within a subject taking place on consecutive days (constraint (3.26)) or exams not taking place at the weekend (constraint (3.20)), the Monday before would also have no exams on it.

Two additional limitations relate to elements of the objective function. The first is how the $W_{i,j}$ values are decided. For this study, they were based on literature on similar studies. However, this might not be fully appropriate to the IB context, given the specific complexities of the IB's situation. Given that the overall goal is to minimise the stress for students caused by the exam schedule, an alternative approach could be to survey students or schools and ask for their views on the extent to which the different proximities of pairs of exams cause issues for them in a measurable way.

Given that the IB is committed to publishing the examination timetable before students are registered for their examinations, the $C_{i,j}$ values are based on the most recent year's data available and as such are only estimates of the true $C_{i,j}$ values in the examination session. Therefore, an extension to this study could be to apply robust optimisation techniques to account for this.

7.2 Future work

The creation of a model that is proven to work on toy datasets opens up a number of potential avenues of future work and investigation, in addition to those that address the limitations in the previous section.

Further amendments to the model

The most important element of future work in this area is to further amend the model or the way it is solved to make it more operationally feasible. This was discussed at length in Section 6, which showed that ordering exams within a subject had a considerable positive impact on computation time. One option could be to further constrain when exams could take place by assigning certain sub-groups of subjects to pre-determined sections of the examination window. For instance, assigning all the individuals and societies subjects to the first full week of exams and all the science subjects to the third week.

Impact of different elements on the objective value

Once a final model that is feasible to run has been created, it opens up a number of different areas of investigation into the impact of different elements of the model on the overall spread of exams. A key area for investigation could be the length of the examination window. For instance, if it was extended by starting earlier or finishing later, would it have a big impact on the overall spread of exams for students? Conversely, could a shorter examination window be introduced without a major impact on the objective function?

Consideration of Fairness

One area that has not been considered within this study is that of whether the optimised exam timetable could be considered to be 'fair'. The optimal solution would result maximising the spread of exams for all students collectively by minimising the objective function. However, how even that spread is across students, or groups of students is not known. This is particularly important if there is some form of bias against a certain demographic of students. There are a number of different mathematical formulations of how to incorporate fairness within an optimisation model (Xinying Chen & Hooker, 2023) and generally these involve

creating student-level utility scores for all students taking exams in an exam session. Given the computational challenges found throughout this study, adding such a complex additional layer to the model would likely be unfeasible, so an alternative approach would be needed. One possibility would be to add constraints relating to popular 'routes' through the Diploma or to conduct post-hoc checks of the utility scores of particular groups of students.

Production of an operational tool

Perhaps the most important additional work is to produce a tool that the IB can use operationally. Producing examination timetables automatically that provide a good spread across the timetable for as many students as possible will not only save time and effort but will improve performance and reduce stress for students. Furthermore, being able to model the impact of proposed changes in policy and rules in this area on the overall quality of timetables will provide huge opportunities for the IB carry out future changes in a safe and informed manner.

7.3 Summary

This study presents a real-world examination timetabling problem in a large-scale international awarding organisation, providing a very different context from many published studies from a university setting and so adds to the already well-established literature in this area. It provides the mixed integer linear optimisation formulation of the model, with the objective of maximising the spread of examinations for students whilst complying to all the rules.

The model is able to solve cases with a small, synthetic dataset of 20 exams over 16 days. However, it failed to provide the optimal solution within a reasonable timeframe when using real-world IB data, which totalled 64 exams over 23 days, requiring manual termination after over 10 hours. Possible ways to improve computation time are provided and it showed that ordering exams within a subject cut the number of work units needed to solve the toy model to 3.55% of the original model, which drops to 1.93% when combined with a simplified model with a reduced objective function. However, even this still proved too burdensome. Further consideration will need to be given as to how to best produce exam timetables given this finding; whether the optimal solution to a changed model is preferred to a good, but not necessarily optimal, solution to the model in this study.

References

AlainChabrier. (2021, May 5). Warm Start Optimization. *Medium*. https://medium.com/@AlainChabrier/warm-start-optimization-ac73eef189e9

Bazari, S., Pooya, A., Soleimani Fard, O., & Roozkhosh, P. (2023). Modeling and solving the problem of scheduling university exams in terms of new constraints on the conflicts of professors' exams and the concurrence of exams with common questions. *OPSEARCH*, *60*(2), 877–915. https://doi.org/10.1007/s12597-023-00638-z

Carter, M. W., Laporte, G., & Lee, S. Y. (1996). Examination Timetabling: Algorithmic Strategies and Applications. *Journal of the Operational Research Society*, *47*(3), 373–383. https://doi.org/10.1057/jors.1996.37

IBO. (n.d.). *IB Diploma Programme and IB Career-related Programme—May 2024 examination schedule*. Retrieved July 29, 2024, from https://www.ibo.org/globalassets/new-structure/programmes/dp/pdfs/may-2024-examination-schedule-exam-en.pdf

International Baccalaureate Organisation. (2017). *IB Diploma Programme Assessment Guidance*. International Bacclaureate.

Jiménez-Cordero, A., Morales, J. M., & Pineda, S. (2022). Warm-starting constraint generation for mixed-integer optimization: A Machine Learning approach. *Knowledge-Based Systems*, *253*, 109570. https://doi.org/10.1016/j.knosys.2022.109570

McCollum, B., McMullan, P., Parkes, A. J., Burke, E. K., & Qu, R. (2012). A new model for automated examination timetabling. *Annals of Operations Research*, *194*(1), 291–315. https://doi.org/10.1007/s10479-011-0997-x

McCollum, B., McMullan, P., Parkes, A. J., & Qu, R. (2007). *The Second International Timetabling Competition: Examination Timetabling Track*.

MIPGap. (n.d.). *Gurobi Optimization*. Retrieved July 31, 2024, from https://www.gurobi.com/documentation/current/refman/mipgap2.html

Mohmad Kahar, M. N., & Kendall, G. (2014). Universiti Malaysia Pahang examination timetabling problem: Scheduling invigilators. *Journal of the Operational Research Society*, *65*(2), 214–226. https://doi.org/10.1057/jors.2012.6

Muklason, A., Parkes, A. J., Özcan, E., McCollum, B., & McMullan, P. (2017). Fairness in examination timetabling: Student preferences and extended formulations. *Applied Soft Computing*, *55*, 302–318. https://doi.org/10.1016/j.asoc.2017.01.026

Müller, T. (2016). Real-life examination timetabling. *Journal of Scheduling*, *19*(3), 257–270. https://doi.org/10.1007/s10951-014-0391-z

Newton, P. E. (2012). Making sense of decades of debate on inter-subject comparability in England. *Assessment in Education: Principles, Policy & Practice*, *19*(2), 251–273. https://doi.org/10.1080/0969594X.2011.563357

Nisbet, I., & Shaw, S. D. (2019). Fair assessment viewed through the lenses of measurement theory. *Assessment in Education: Principles, Policy & Practice*, *26*(5), 612–629. https://doi.org/10.1080/0969594X.2019.1586643

Organization, I. B. (2014, August 22). *DP curriculum*. International Baccalaureate®. https://www.ibo.org/programmes/diploma-programme/curriculum/

Ozcan, E., & Ersoy, E. (2005). Final Exam Scheduler—FES. *2005 IEEE Congress on Evolutionary Computation*, *2*, 1356–1363. https://doi.org/10.1109/CEC.2005.1554848

Qu, R., Burke, E. K., McCollum, B., Merlot, L. T. G., & Lee, S. Y. (2009). A survey of search methodologies and automated system development for examination timetabling. *Journal of Scheduling*, *12*(1), 55–89. https://doi.org/10.1007/s10951-008-0077-5

Siew, E. S. K., Sze, S. N., Goh, S. L., Kendall, G., Sabar, N. R., & Abdullah, S. (2024). A Survey of Solution Methodologies for Exam Timetabling Problems. *IEEE Access*, *12*, 41479–41498. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3378054

Work. (n.d.). *Gurobi Optimization*. Retrieved July 31, 2024, from https://www.gurobi.com/documentation/current/refman/work.html

Xinying Chen, V., & Hooker, J. N. (2023). A guide to formulating fairness in an optimization model. *Annals of Operations Research*, *326*(1), 581–619. https://doi.org/10.1007/s10479-023-05264-y

Appendix 1 – Detail of the toy model

Appendix 1.1 – List of exams, subjects and exam lengths

Subject	Exam	Full Name	Length (hrs)	Exam number
Optimisation	Exam 1	Optimisation Exam 1	2	1
Optimisation	Exam 2	Optimisation Exam 2	2.5	2
Statistics	Exam 1	Statistics Exam 1	3.5	3
Statistics	Exam 2	Statistics Exam 2	1.5	4
SBS	Exam 1	SBS Exam 1	2	5
SBS	Exam 2	SBS Exam 2	2	6
Language A Literature	Exam 1	Language A Literature Exam 1	2	7
Language A Literature	Exam 2	Language A Literature Exam 2	2	8
Machine Learning	Exam 1	Machine Learning Exam 1	1	9
Machine Learning	Exam 2	Machine Learning Exam 2	2.25	10
Deep Learning	Exam 1	Deep Learning Exam 1	1.5	11
Deep Learning	Exam 2	Deep Learning Exam 2	3.5	12
Coding Lab	Exam 1	Coding Lab Exam 1	2.5	13
Coding Lab	Exam 2	Coding Lab Exam 2	2	14
Data Stewardship	Exam 1	Data Stewardship Exam 1	2.25	15
Data Stewardship	Exam 2	Data Stewardship Exam 2	2	16
Finance	Exam 1	Finance Exam 1	2	17
Finance	Exam 2	Finance Exam 2	2	18
Law and Ethics	Exam 1	Law and Ethics Exam 1	1.75	19
Law and Ethics	Exam 2	Law and Ethics Exam 2	3	20

Appendix 1.2 – Matrix of $C_{i,j}$ values

Exam numbers correspond to the exams given in Appendix 1.1

Exam No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	50	50	25	25	30	30	35	35	10	10	15	15	30	30	40	40	5	5	0	0
2	50	50	25	25	30	30	35	35	10	10	15	15	30	30	40	40	5	5	0	0
3	25	25	50	50	0	0	5	5	35	35	30	30	20	20	45	45	10	10	10	10
4	25	25	50	50	0	0	5	5	35	35	30	30	20	20	45	45	10	10	10	10
5	30	30	0	0	50	50	40	40	10	10	15	15	20	20	10	10	25	25	35	35
6	30	30	0	0	50	50	40	40	10	10	15	15	20	20	10	10	25	25	35	35
7	35	35	5	5	40	40	50	50	15	15	20	20	10	10	15	15	5	5	45	45
8	35	35	5	5	40	40	50	50	15	15	20	20	10	10	15	15	5	5	45	45
9	10	10	35	35	10	10	15	15	50	50	45	45	45	45	30	30	20	20	15	15
10	10	10	35	35	10	10	15	15	50	50	45	45	45	45	30	30	20	20	15	15
11	15	15	30	30	15	15	20	20	45	45	50	50	40	40	25	25	25	25	20	20
12	15	15	30	30	15	15	20	20	45	45	50	50	40	40	25	25	25	25	20	20
13	30	30	20	20	20	20	10	10	45	45	40	40	50	50	30	30	20	20	10	10
14	30	30	20	20	20	20	10	10	45	45	40	40	50	50	30	30	20	20	10	10
15	40	40	45	45	10	10	15	15	30	30	25	25	30	30	50	50	30	30	20	20
16	40	40	45	45	10	10	15	15	30	30	25	25	30	30	50	50	30	30	20	20
17	5	5	10	10	25	25	5	5	20	20	25	25	20	20	30	30	50	50	35	35
18	5	5	10	10	25	25	5	5	20	20	25	25	20	20	30	30	50	50	35	35
19	0	0	10	10	35	35	45	45	15	15	20	20	10	10	20	20	35	35	50	50
20	0	0	10	10	35	35	45	45	15	15	20	20	10	10	20	20	35	35	50	50

Appendix 1.3 – Python code for the toy model

```
#install relevant libraries
import numpy as np
from gurobipy import *
import pandas as pd

#read in the toy data
exams2=pd.read_csv("Toy exam list.csv")
#pairs=pd.read_csv("Exam pairs.csv")
pairs=pd.read_csv("Exam pairs.csv",index_col=0)
days=pd.read_csv("exam_days3.csv")


#create full date
days['Full_Date']=days["DOW"] + " " + days["Date"]


#####################################################################
################
#############################
#############################    Hyperparameters
#############################
#####################################################################
################

# Number of possible exams
#nb_exam =len(exams)
nb_exam=20
# Number of days - we'll say 16 days for now
nb_day = 16
# Number of slots per day - AM is 1, PM is 2
nb_slot = 2

#maximum number of exam clashes
max_clashes=15

# Define penalties here

# same slot penalty
W_a = 64
# same day penalty
W_b = 32
#next day penalty
W_c = 16
#one day gap penalty
W_d = 8
#two day gap penalty
W_e = 4
#three day gap penalty
W_f = 2
#four day gap penalty
W_g = 1
# TODO: define other penalties below: W's, C's, L's

#create exam lengths
exam_length=exams2['Length']
```

```
#define common student matrix

C=np.array(pairs)


#subset list of dates to the right length and days
days2=days.head(nb_day)

exams=exams2.head(nb_exam)
#####create some lists to take them out of constraints
##weekend days
weekend_days=days2[days2['DOW'].isin(["Sat","Sun"])].index.tolist()
##May 1st
May1=days2[days2['Date'] == "01/05/2026"].index.tolist()


#################################################################
################
#############################
#############################    Optimization model
#############################
#################################################################
################

# Initialize the optimization model
mod = Model()
    # Choose to output the feedback from the optimization model
mod.setParam('OutputFlag', 1)

########### Defining all variables

### First define arrays of None
# x[i,p,m]: assigning exam i, slot p, day m
x=np.array([[[None for day in range(nb_day)] for slot in
range(nb_slot)] for i in range(nb_exam)])

# y_a[i,j]: indicating exams i,j are assigned to same slots
y_a=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_b=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_c=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_d=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_e=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_f=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])
y_g=np.array([[None for j in range(nb_exam)] for j in
range(nb_exam)])

#set-up all the y variables
for i in range(nb_exam):
  for j in range(nb_exam):
    if i<j:#this should ensure i<j
```

```
        y_a[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_a
        y_b[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_b
        y_c[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_c
        y_d[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_d
        y_e[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_e
        y_f[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_f
        y_g[i,j]=mod.addVar(vtype=GRB.CONTINUOUS,lb=0.0,ub=1.0) #y_f

#set-up xipm
for i in range(nb_exam):
    for m in range(nb_day):
        for p in range(nb_slot):
            x[i,p,m]=mod.addVar(vtype=GRB.BINARY) #x[i,p,m]

# Updating the model with the new variables
mod.update()

########### Defining the objective function
objectif = LinExpr()
for i in range(nb_exam):
  for j in range(nb_exam):
    if i<j:

      objectif+=C[i,j]*(W_a*y_a[i,j]+W_b*y_b[i,j]+W_c*y_c[i,j]+W_d*y_
      d[i,j]+W_e*y_e[i,j]+W_f*y_f[i,j]+W_g*y_g[i,j])

mod.setObjective(objectif, GRB.MINIMIZE)
########### Defining constraints

# linking y_a and x:
for i in range(nb_exam):
  for j in range(nb_exam):
    for p in range(nb_slot):
      for m in range(nb_day):
        if i<j:
            mod.addConstr(y_a[i,j]>=x[i,p,m]+x[j,p,m]-1)

# linking y_b and x:
for i in range(nb_exam):
  for j in range(nb_exam):
      for m in range(nb_day):
        if i<j:
         mod.addConstr(y_b[i,j]>=x[i,0,m]+x[j,0,m]+x[i,1,m]+x[j,1,m]-
1)

# linking y_c and x:
for i in range(nb_exam):
  for j in range(nb_exam):
      for m in range(nb_day-1):
        if i<j:

            mod.addConstr(y_c[i,j]>=x[i,0,m]+x[j,0,m+1]+x[i,1,m]+x[j,
            1,m+1]-1) #forwards

            mod.addConstr(y_c[i,j]>=x[i,0,m+1]+x[j,0,m]+x[i,1,m+1]+x[
            j,1,m]-1) #backwards
```

```
#linking y_d and x :
for i in range(nb_exam):
  for j in range(nb_exam):
      for m in range(nb_day-2):
        if i<j:

            mod.addConstr(y_d[i,j]>=x[i,0,m]+x[j,0,m+2]+x[i,1,m]+x[j,
            1,m+2]-1) #forwards

            mod.addConstr(y_d[i,j]>=x[i,0,m+2]+x[j,0,m]+x[i,1,m+2]+x[
            j,1,m]-1) #backwards


# linking y_e and x:
for i in range(nb_exam):
  for j in range(nb_exam):
      for m in range(nb_day-3):
        if i<j:

            mod.addConstr(y_e[i,j]>=x[i,0,m]+x[j,0,m+3]+x[i,1,m]+x[j,
            1,m+3]-1) #forwards

            mod.addConstr(y_e[i,j]>=x[i,0,m+3]+x[j,0,m]+x[i,1,m+3]+x[
            j,1,m]-1) #backwards

# linking y_f and x:
for i in range(nb_exam):
    for j in range(nb_exam):
      for m in range(nb_day-4):
        if i<j:
            mod.addConstr(y_f[i,j]>=x[i,0,m]+x[j,0,m+4]+x[i,1,m]+x[j,
            1,m+4]-1) #forwards
            mod.addConstr(y_f[i,j]>=x[i,0,m+4]+x[j,0,m]+x[i,1,m+4]+x[
            j,1,m]-1) #backwards

# linking y_g and x:
for i in range(nb_exam):
  for j in range(nb_exam):
      for m in range(nb_day-5):
        if i<j:
            mod.addConstr(y_c[i,j]>=x[i,0,m]+x[j,0,m+5]+x[i,1,m]+x[j,
            1,m+5]-1) #forwards
            mod.addConstr(y_c[i,j]>=x[i,0,m+5]+x[j,0,m]+x[i,1,m+5]+x[
            j,1,m]-1) #backwards


#each exam gets exactly one slot
for i in range(nb_exam):
  mod.addConstr(quicksum(x[i,p,m] for p in range(nb_slot) for m in
range(nb_day))==1)

#no exams at a weekend
for i in range(nb_exam):
  for p in range(nb_slot):
    for m in weekend_days:
        mod.addConstr(x[i,p,m]==0)
```

```
#No lang lit on a Friday
for i in exams[exams['Subject'] == "Language A
Literature"].index.tolist():
  for p in range(nb_slot):
    for m in days2[days2['DOW'] == 'Fri'].index.tolist():
        mod.addConstr(x[i,p,m]==0)

#No exams on May 1st
for i in range(nb_exam):
  for p in range(nb_slot):
    for m in May1:
        mod.addConstr(x[i,p,m]==0)

#keep number of exam clashes below a certain number
mod.addConstr(quicksum(C[i,j]*y_a[i,j] for i in range(nb_exam) for j
in range(nb_exam) if i<j) <=max_clashes)

#add the maximum number of hours in the afternoon constraint
for i in range(nb_exam):
  for m in range(nb_day):
     if exams.loc[i,'Length']>3:
            mod.addConstr(x[i,1,m]==0)

#No more than 6.25 hours exams in a day
for i in range(nb_exam):
    for j in range(nb_exam):
        if i<j:
            if exams.loc[i,'Length'] + exams.loc[j,'Length'] > 6.25:
                mod.addConstr(y_b[i,j]==0)

#SBSs start on day 1
for i in exams[exams['Full Name'] == "SBS Exam 1"].index.tolist():
    mod.addConstr(x[i,0,0]==1)

#All other exams not on day 1
for i in exams[exams['Subject'] != "SBS"].index.tolist():
  for p in range(nb_slot):
    mod.addConstr(x[i,p,0]+x[i,p,1]==0)

#exams in same subject on consecutive days
for i in range(nb_exam):
    for j in range(nb_exam):
        if i<j:
            if exams.loc[i,'Subject']==exams.loc[j,'Subject']:
                for p in range(nb_slot):
                    for m in range(nb_day):
                       mod.addConstr(x[j,p,m]+(sum(x[i,q,n] for q in
range(nb_slot) for n in range(nb_day) if n not in (m-1,m+1)))<=1)

#put finance and law and ethics in the first half

for i in exams[exams['Subject'].isin(["Finance","Law and
Ethics"])].index.tolist():
    mod.addConstr(quicksum(x[i,p,m] for p in range(nb_slot) for m in
range(round(nb_day/2),nb_day,1))==0)
```

Appendix 2 – Recoding of IB exams

Recoding of subjects

Old Name	Recoded Name
ECONOMICS P11	ECONOMICS
ECONOMICS P22	ECONOMICS
ENG P A LIT	ENGLISH A LIT
SPA P A LIT	SPANISH A LIT
LANGUAGE B	LANGUAGE ACQUISITION
LANGUAGE AB INITIO	LANGUAGE ACQUISITION
FRENCH B	FRENCH LANGUAGE ACQUISITION
FRENCH AB.	FRENCH LANGUAGE ACQUISITION
ENGLISH B	ENGLISH LANGUAGE ACQUISITION
ENGLISH AB	ENGLISH LANGUAGE ACQUISITION
SPANISH B	SPANISH LANGUAGE ACQUISITION
SPANISH AB	SPANISH LANGUAGE ACQUISITION
PSYCHOLOGY P11	PSYCHOLOGY
PSYCHOLOGY P22	PSYCHOLOGY

Recoding and combining of exams

Old Name	Recoded Name
BIOLOGY PAPER 1 (MCQ)	BIOLOGY P1 AND P3
BIOLOGY PAPER THREE	BIOLOGY P1 AND P3
BUSINESS MANAGEMENT PAPER ONE	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
BUSINESS MANAGEMENT PAPER THREE	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
CHEMISTRY PAPER 1 (MCQ)	CHEMISTRY P1 AND P3
CHEMISTRY PAPER THREE	CHEMISTRY P1 AND P3
COMPUTER SC. PAPER THREE	COMPUTER SC. PAPER TWO AND HL PAPER THREE
COMPUTER SC. PAPER TWO	COMPUTER SC. PAPER TWO AND HL PAPER THREE
DESIGN TECH. PAPER 1 (MCQ)	DES TECH P1 AND P2
DESIGN TECH. PAPER TWO	DES TECH P1 AND P2

DIGITAL SOC. PAPER THREE	DIGITAL SOC. PAPER TWO AND PAPER THREE
DIGITAL SOC. PAPER TWO	DIGITAL SOC. PAPER TWO AND PAPER THREE
ECONOMICS PAPER THREE	ECONOMICS PAPER TWO AND PAPER THREE
ECONOMICS PAPER TWO	ECONOMICS PAPER TWO AND PAPER THREE
ENGLISH A LIT PAPER TWO	ENGLISH A LAL PAPER TWO
FRENCH A LIT PAPER TWO	FRENCH A LAL PAPER TWO
GEOGRAPHY PAPER THREE	GEOGRAPHY PAPER TWO AND PAPER THREE
GEOGRAPHY PAPER TWO	GEOGRAPHY PAPER TWO AND PAPER THREE
LANGUAGE A LITERATURE PAPER TWO	LANGUAGE A LANGUAGE AND LITERATURE PAPER TWO
LANGUAGE ACQUISITION PAPER ONE	LANGUAGE ACQUISITION P1 and P2 READING
LANGUAGE ACQUISITION PAPER TWO READING	LANGUAGE ACQUISITION P1 and P2 READING
LITERATURE AND PERFORMANCE PAPER ONE	ENGLISH A LAL PAPER TWO
PHILOSOPHY PAPER THREE	PHILOSOPHY PAPER TWO AND HL PAPER THREE
PHILOSOPHY PAPER TWO	PHILOSOPHY PAPER TWO AND HL PAPER THREE
PHYSICS PAPER 1 (MCQ)	PHYSICS P1 AND P3
PHYSICS PAPER THREE	PHYSICS P1 AND P3
PSYCHOLOGY PAPER THREE	PSYCHOLOGY PAPER TWO AND PAPER THREE
PSYCHOLOGY PAPER TWO	PSYCHOLOGY PAPER TWO AND PAPER THREE
SPANISH A LIT PAPER TWO	SPANISH A LAL PAPER TWO
SPORTS EX SCI PAPER 1 (MCQ)	SPORTS EX SCI P1 AND P3
SPORTS EX SCI PAPER THREE	SPORTS EX SCI P1 AND P3
BIOLOGY PAPER 1 (MCQ)	BIOLOGY P1 AND P3

BIOLOGY PAPER THREE	BIOLOGY P1 AND P3
BUSINESS MANAGEMENT PAPER ONE	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
BUSINESS MANAGEMENT PAPER THREE	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
CHEMISTRY PAPER 1 (MCQ)	CHEMISTRY P1 AND P3
CHEMISTRY PAPER THREE	CHEMISTRY P1 AND P3
COMPUTER SC. PAPER THREE	COMPUTER SC. PAPER TWO AND HL PAPER THREE
COMPUTER SC. PAPER TWO	COMPUTER SC. PAPER TWO AND HL PAPER THREE

Appendix 3 – IB exam timetable at the point of manual termination

Day of Week	Date	Slot	Exam Name
Thursday	23/04/2026	AM	SBS PAPER ONE
Friday	24/04/2026	AM	SBS PAPER TWO
Monday	27/04/2026	AM	ENGLISH A LAL PAPER ONE
Monday	27/04/2026	AM	ENGLISH LANGUAGE ACQUISITION PAPER TWO LISTENING
Monday	27/04/2026	PM	MATHEMATICS ANALYSIS AND APPROACHES PAPER THREE
Monday	27/04/2026	PM	MATHEMATICS APPLICATIONS AND INTERP PAPER THREE
Tuesday	28/04/2026	AM	ENGLISH ACQUISITION P1 and P2 READING
Tuesday	28/04/2026	PM	ENGLISH A LAL PAPER TWO
Wednesday	29/04/2026	AM	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
Wednesday	29/04/2026	AM	SOC.CUL.ANTH. PAPER TWO
Wednesday	29/04/2026	PM	HISTORY PAPER THREE
Thursday	30/04/2026	AM	BUSINESS MANAGEMENT PAPER TWO
Thursday	30/04/2026	AM	SOC.CUL.ANTH. PAPER ONE
Thursday	30/04/2026	PM	HISTORY PAPER ONE AND PAPER TWO
Monday	04/05/2026	AM	FRENCH A LAL PAPER ONE
Monday	04/05/2026	AM	LANGUAGE A LANGUAGE AND LITERATURE PAPER TWO
Monday	04/05/2026	AM	LATIN PAPER TWO
Monday	04/05/2026	AM	SPANISH A LAL PAPER ONE
Monday	04/05/2026	PM	CLASS.GREEK PAPER ONE
Monday	04/05/2026	PM	FRENCH LANGUAGE ACQUISITION PAPER TWO LISTENING
Monday	04/05/2026	PM	LANGUAGE ACQUISITION P1 and P2 READING
Monday	04/05/2026	PM	SPANISH ACQUISITION P1 and P2 READING
Tuesday	05/05/2026	AM	FRENCH ACQUISITION P1 and P2 READING
Tuesday	05/05/2026	AM	LANGUAGE ACQUISITION PAPER TWO LISTENING

Tuesday	05/05/2026	AM	LATIN PAPER ONE
Tuesday	05/05/2026	AM	SPANISH LANGUAGE ACQUISITION PAPER TWO LISTENING
Tuesday	05/05/2026	PM	CLASS.GREEK PAPER TWO
Tuesday	05/05/2026	PM	FRENCH A LAL PAPER TWO
Tuesday	05/05/2026	PM	LANGUAGE A LANGUAGE AND LITERATURE PAPER ONE
Tuesday	05/05/2026	PM	SPANISH A LAL PAPER TWO
Tuesday	05/05/2026	PM	WORLD RELIG. PAPER ONE
Wednesday	06/05/2026	AM	WORLD RELIG. PAPER TWO
Thursday	07/05/2026	AM	CHEMISTRY PAPER TWO
Thursday	07/05/2026	AM	COMPUTER SC. PAPER TWO AND HL PAPER THREE
Thursday	07/05/2026	AM	DESIGN TECH. PAPER THREE
Thursday	07/05/2026	PM	ECONOMICS PAPER TWO AND PAPER THREE
Thursday	07/05/2026	PM	PHILOSOPHY PAPER ONE
Thursday	07/05/2026	PM	PSYCHOLOGY PAPER ONE
Friday	08/05/2026	AM	ECONOMICS PAPER ONE
Friday	08/05/2026	AM	PHILOSOPHY PAPER TWO AND HL PAPER THREE
Friday	08/05/2026	AM	PSYCHOLOGY PAPER TWO AND PAPER THREE
Friday	08/05/2026	PM	CHEMISTRY P1 AND P3
Friday	08/05/2026	PM	COMPUTER SC. PAPER ONE
Friday	08/05/2026	PM	DES TECH P1 AND P2
Monday	11/05/2026	AM	BIOLOGY PAPER TWO
Monday	11/05/2026	PM	ENV. AND SOC. PAPER TWO
Monday	11/05/2026	PM	NATURE OF SCIENCE PAPER TWO
Monday	11/05/2026	PM	PHYSICS PAPER TWO
Monday	11/05/2026	PM	SPORTS EX SCI PAPER TWO
Tuesday	12/05/2026	AM	BIOLOGY P1 AND P3
Tuesday	12/05/2026	PM	ENV. AND SOC. PAPER ONE
Tuesday	12/05/2026	PM	NATURE OF SCIENCE PAPER ONE
Tuesday	12/05/2026	PM	PHYSICS P1 AND P3
Tuesday	12/05/2026	PM	SPORTS EX SCI P1 AND P3
Wednesday	13/05/2026	AM	DIGITAL SOC. PAPER TWO AND PAPER THREE
Wednesday	13/05/2026	PM	GLOBAL POLITICS PAPER ONE

Thursday	14/05/2026	AM	MATHEMATICS ANALYSIS AND APPROACHES PAPER ONE
Thursday	14/05/2026	AM	MATHEMATICS APPLICATIONS AND INTERP PAPER ONE
Thursday	14/05/2026	PM	DIGITAL SOC. PAPER ONE
Thursday	14/05/2026	PM	GEOGRAPHY PAPER TWO AND PAPER THREE
Thursday	14/05/2026	PM	GLOBAL POLITICS PAPER TWO
Friday	15/05/2026	AM	GEOGRAPHY PAPER ONE
Friday	15/05/2026	PM	MATHEMATICS ANALYSIS AND APPROACHES PAPER TWO
Friday	15/05/2026	PM	MATHEMATICS APPLICATIONS AND INTERP PAPER TWO

Appendix 4 – IB timetable at point of termination with amended model

Day of Week	Date	Slot	Exam Name
Thursday	23/04/2026	AM	SBS PAPER ONE
Friday	24/04/2026	AM	SBS PAPER TWO
Thursday	23/04/2026	AM	SBS PAPER ONE
Friday	24/04/2026	AM	SBS PAPER TWO
Monday	27/04/2026	AM	GEOGRAPHY PAPER ONE
Monday	27/04/2026	AM	HISTORY PAPER ONE AND PAPER TWO
Monday	27/04/2026	PM	BUSINESS MANAGEMENT PAPER ONE AND HL PAPER THREE
Monday	27/04/2026	PM	SOC.CUL.ANTH. PAPER ONE
Tuesday	28/04/2026	AM	BUSINESS MANAGEMENT PAPER TWO
Tuesday	28/04/2026	AM	SOC.CUL.ANTH. PAPER TWO
Tuesday	28/04/2026	PM	GEOGRAPHY PAPER TWO AND PAPER THREE
Tuesday	28/04/2026	PM	HISTORY PAPER THREE
Wednesday	29/04/2026	PM	ENGLISH A LAL PAPER ONE
Wednesday	29/04/2026	PM	ENGLISH ACQUISITION P1 and P2 READING
Thursday	30/04/2026	AM	ENGLISH A LAL PAPER TWO
Thursday	30/04/2026	PM	ENGLISH LANGUAGE ACQUISITION PAPER TWO LISTENING
Monday	04/05/2026	AM	CHEMISTRY P1 AND P3
Monday	04/05/2026	AM	DES TECH P1 AND P2
Monday	04/05/2026	PM	ECONOMICS PAPER ONE
Monday	04/05/2026	PM	PSYCHOLOGY PAPER ONE
Tuesday	05/05/2026	AM	ECONOMICS PAPER TWO AND PAPER THREE
Tuesday	05/05/2026	AM	PSYCHOLOGY PAPER TWO AND PAPER THREE
Tuesday	05/05/2026	AM	WORLD RELIG. PAPER ONE
Tuesday	05/05/2026	PM	CHEMISTRY PAPER TWO
Tuesday	05/05/2026	PM	COMPUTER SC. PAPER ONE
Tuesday	05/05/2026	PM	DESIGN TECH. PAPER THREE
Wednesday	06/05/2026	AM	DIGITAL SOC. PAPER ONE
Wednesday	06/05/2026	AM	GLOBAL POLITICS PAPER ONE
Wednesday	06/05/2026	AM	PHILOSOPHY PAPER ONE

Wednesday	06/05/2026	PM	COMPUTER SC. PAPER TWO AND HL PAPER THREE
Wednesday	06/05/2026	PM	WORLD RELIG. PAPER TWO
Thursday	07/05/2026	AM	MATHEMATICS ANALYSIS AND APPROACHES PAPER ONE
Thursday	07/05/2026	AM	MATHEMATICS APPLICATIONS AND INTERP PAPER ONE
Thursday	07/05/2026	PM	DIGITAL SOC. PAPER TWO AND PAPER THREE
Thursday	07/05/2026	PM	GLOBAL POLITICS PAPER TWO
Thursday	07/05/2026	PM	PHILOSOPHY PAPER TWO AND HL PAPER THREE
Friday	08/05/2026	AM	MATHEMATICS APPLICATIONS AND INTERP PAPER TWO
Friday	08/05/2026	PM	MATHEMATICS ANALYSIS AND APPROACHES PAPER TWO
Monday	11/05/2026	AM	FRENCH A LAL PAPER ONE
Monday	11/05/2026	AM	LANGUAGE A LANGUAGE AND LITERATURE PAPER ONE
Monday	11/05/2026	AM	LATIN PAPER ONE
Monday	11/05/2026	AM	SPANISH A LAL PAPER ONE
Monday	11/05/2026	PM	CLASS.GREEK PAPER ONE
Monday	11/05/2026	PM	FRENCH ACQUISITION P1 and P2 READING
Monday	11/05/2026	PM	LANGUAGE ACQUISITION P1 and P2 READING
Monday	11/05/2026	PM	SPANISH ACQUISITION P1 and P2 READING
Tuesday	12/05/2026	AM	CLASS.GREEK PAPER TWO
Tuesday	12/05/2026	AM	FRENCH LANGUAGE ACQUISITION PAPER TWO LISTENING
Tuesday	12/05/2026	AM	LANGUAGE ACQUISITION PAPER TWO LISTENING
Tuesday	12/05/2026	AM	SPANISH LANGUAGE ACQUISITION PAPER TWO LISTENING
Tuesday	12/05/2026	PM	FRENCH A LAL PAPER TWO
Tuesday	12/05/2026	PM	LANGUAGE A LANGUAGE AND LITERATURE PAPER TWO
Tuesday	12/05/2026	PM	LATIN PAPER TWO
Tuesday	12/05/2026	PM	SPANISH A LAL PAPER TWO

Wednesday	13/05/2026	PM	MATHEMATICS ANALYSIS AND APPROACHES PAPER THREE
Wednesday	13/05/2026	PM	MATHEMATICS APPLICATIONS AND INTERP PAPER THREE
Thursday	14/05/2026	AM	ENV. AND SOC. PAPER ONE
Thursday	14/05/2026	AM	NATURE OF SCIENCE PAPER ONE
Thursday	14/05/2026	AM	PHYSICS P1 AND P3
Thursday	14/05/2026	AM	SPORTS EX SCI P1 AND P3
Thursday	14/05/2026	PM	BIOLOGY P1 AND P3
Friday	15/05/2026	AM	ENV. AND SOC. PAPER TWO
Friday	15/05/2026	AM	NATURE OF SCIENCE PAPER TWO
Friday	15/05/2026	AM	PHYSICS PAPER TWO
Friday	15/05/2026	AM	SPORTS EX SCI PAPER TWO
Friday	15/05/2026	PM	BIOLOGY PAPER TWO